Medical Image Compression using SVD

Linear Algebra Mini Project (UE24MA241B) | Department of Computer Science and Engineering | PES University

Team Members

Abhinav Agraharam: PES1UG24CS001
Aadhavan Muthusamy: PES1UG24CS002
Aakash Desai: PES1UG24CS006
Adarsh Rajesh: PES1UG24CS020

Demo Date

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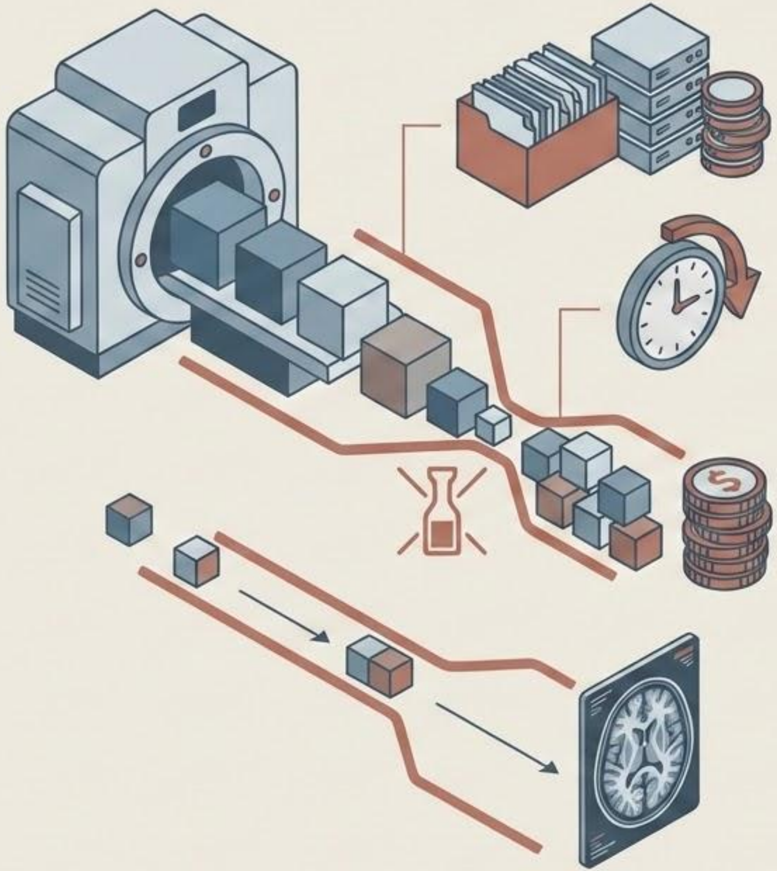


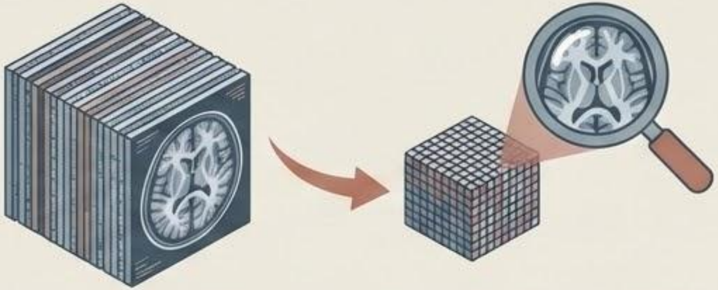
Motivation

- High-resolution medical images (X-ray, CT, MRI) → Large storage requirements
- Sharing scans between hospitals → Slow transmission
- Healthcare systems handle massive data daily → High cost of storage & bandwidth
- Diagnostic quality must be preserved → Compression must be efficient and accurate

Key Idea

Efficient compression can reduce storage and transmission cost while maintaining diagnostic quality

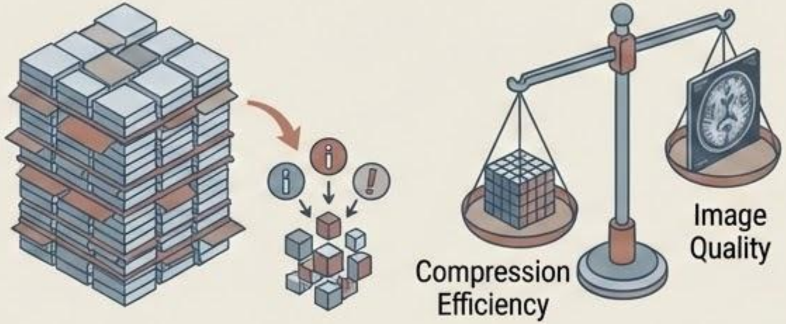
🎯Problem Statement



Goal:

Compress medical images while preserving important diagnostic features.

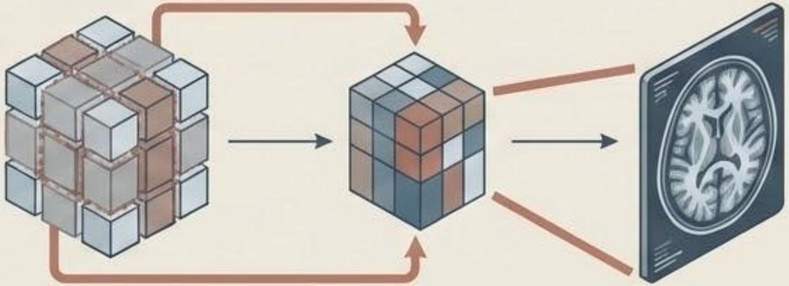
⚠️ Challenges



Images contain large amounts of data
Reducing size may lead to loss of critical information

Need a method that balances:
→ Compression efficiency
→ Image quality

💡 Our Approach

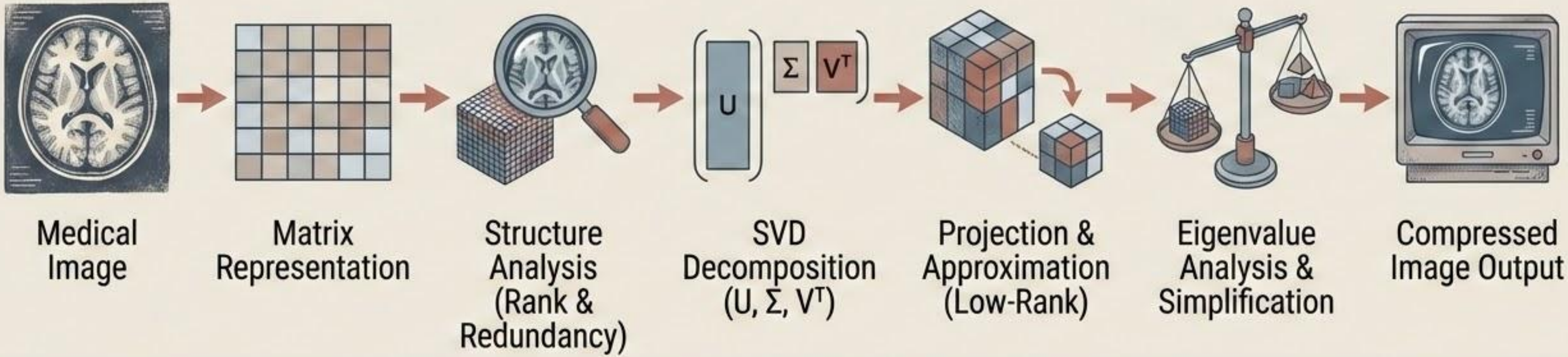


Use Linear Algebra (SVD) to:

- Identify redundant information
- Retain only the most important components
- Reconstruct a high-quality compressed image

Linear Algebra Pipeline

End-to-End Workflow



💡 Key Idea

We transform raw image data into a matrix, apply linear algebra techniques, and reconstruct a compressed version with minimal loss.

Dataset Overview

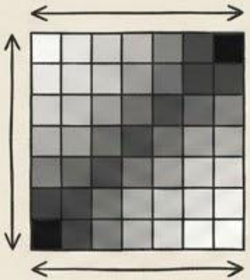
Medical Image Dataset

Chest X-rays

CT Scans

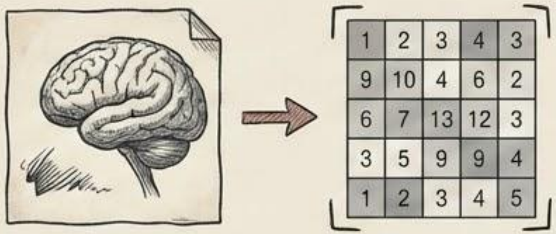
MRI Images

Image Details



Resolution: 512 × 512 pixels

Format: Grayscale



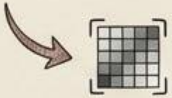
Each image → represented as a matrix of pixel intensities (0–255)

Observation

Images show patterns and similarities.

Indicates redundancy in data.

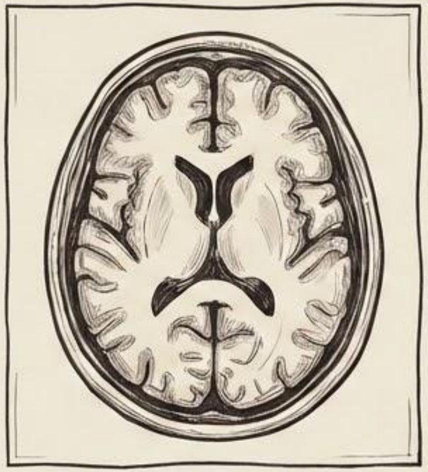
Enables effective compression.



Effective Compression

Matrix Representation

Person 1 — Steps 1–2



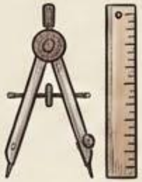
Medical Image
(512x512)

Matrix $A \in \mathbb{R}^{512\times512}$

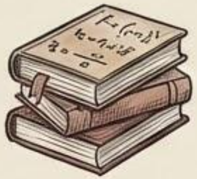
1	2	3	4	...	2	1	4
9	10	4	6	...	8	6	2
6	7	13	12	...	7	3	3
⋮	⋮	9	4	...	⋮	⋮	⋮
3	5	9	4	...	13	9	4
⋮	⋮	⋮	⋮	...	1	2	3
1	2	3	5	...	3	4	5

- Rows → pixel rows
- Columns → pixel columns
- Entries → intensity values (0–255)

Why Matrix Representation?



- Enables use of linear algebra techniques



- Converts real-world data into mathematical form

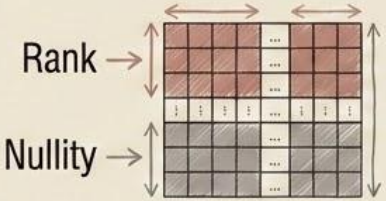
Rank & Redundancy

Person 1 — Step 3

Structure of the Matrix

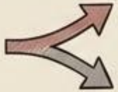
We analyze the matrix using:

- Rank
- Nullity

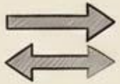


Key Concepts

- **Rank**: Number of independent rows/columns



- **Nullity**: Number of dependent components

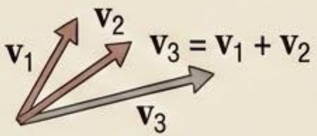


Observation

- Rank is less than full dimension

Rank (2) < Full Dimension (5)

- Indicates linear dependence in data



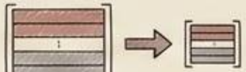
Linear Dependence

Conclusion

- Image contains redundant information

Redundant Data

- This redundancy enables compression



Compressed Data

Singular Value Decomposition

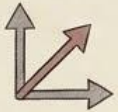
Person 2 — Step 4

SVD Decomposition

$$A = U\Sigma V^T$$

Components

- **U** → Orthogonal basis (row space)

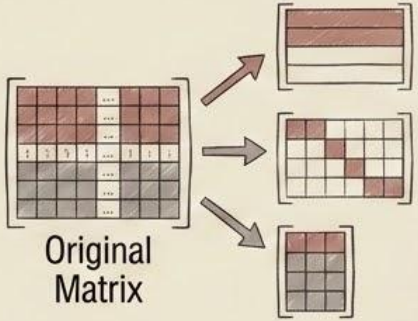


- **Σ** → Singular values (importance of features)

- **V^T** → Orthogonal basis (column space)

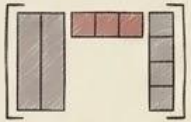
Purpose

- Break matrix into independent components
- Identify important patterns in data

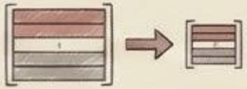


Outcome

- Structured representation of image
- Enables efficient compression



Structured Representation



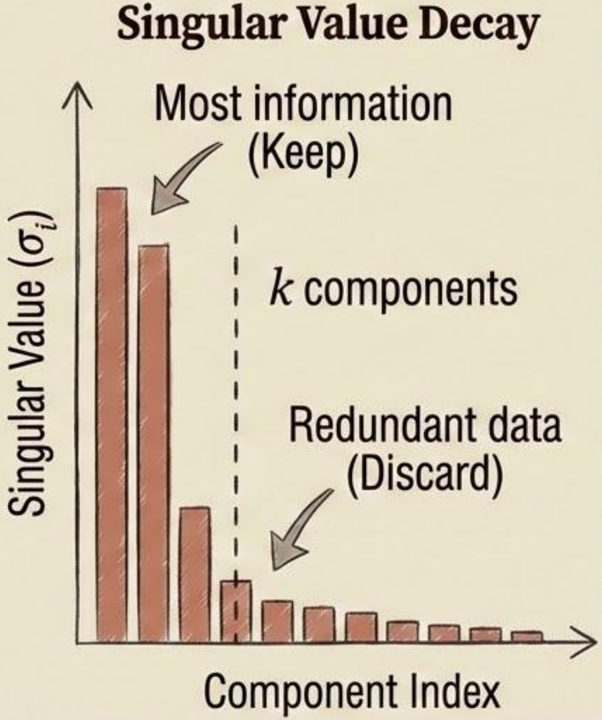
Efficient Compression

Basis Selection (Singular Values)

Person 2 — Step 5

Key Idea

- Singular values indicate importance of components
- Larger values → more important information
- Smaller values → less important details



Observation

- Singular values decay rapidly
- Most information is concentrated in few components

Decision

- Select top k components
- Discard remaining redundant data

Outcome

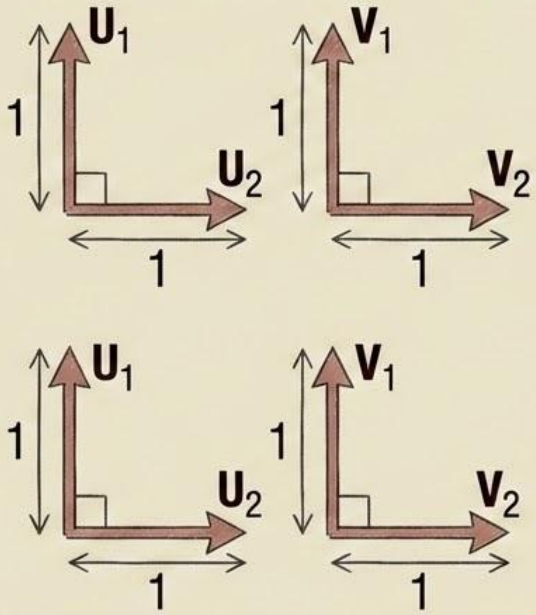
- Reduced dimensionality
- High information retained with fewer components

Orthogonalization

Person 3 — Step 6

Key Idea

- Columns of **U** and rows of $\mathbf{V}^T$ form orthonormal bases
- Vectors are:
  - Independent
  - Normalized



Purpose

- Ensure no overlap (redundancy) between components
- Simplify computations

Outcome

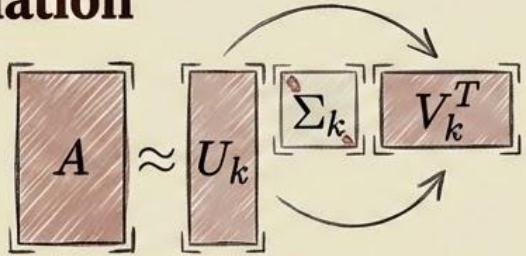
- Stable and meaningful decomposition
- Efficient representation of data

Projection onto Subspace

Person 3 — Step 7

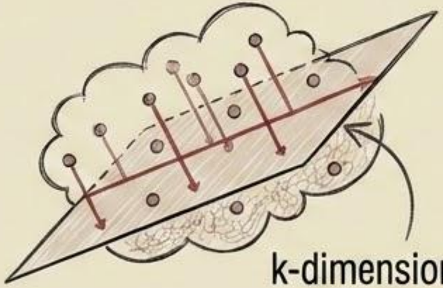
Low-Rank Approximation

$$A_k = U_k \Sigma_k V_k^T$$



Key Idea

- Project the image onto a k-dimensional subspace
- Keep only the most important components



Purpose

- Reduce dimensionality
- Preserve essential structure of the image

Observation

- Small loss of detail
- Most important features retained

Outcome

- Compressed image with minimal error

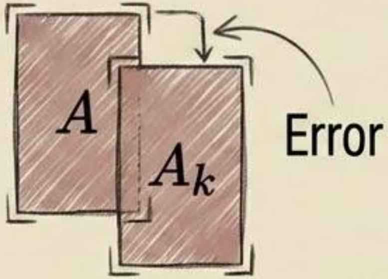
Least Squares Approximation

Person 4 — Step 8

Objective

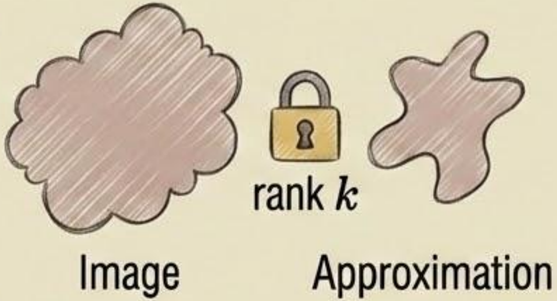
Minimize reconstruction error:

$$\min \|A - A_k\|_F^2$$



Key Idea

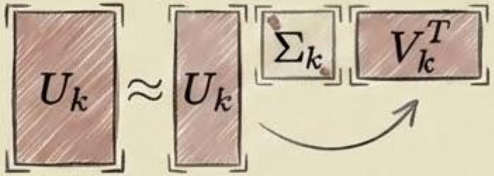
- Find the best possible approximation of the image
- Under the constraint of rank k



Solution

$$A_k = U_k \Sigma_k V^T$$

Obtained directly from SVD



Result

- Minimum possible error
- No other rank-k matrix gives better approximation

Eigenvalue Analysis - Visualization

👤 **Person 4 — Step 9**

🔢 Key Relationship

$\sigma_i^2 = \lambda_i$ of $A^T A$

💡 Key Idea

Singular values are directly related to eigenvalues

Both represent importance of components

🔍 Observation

Large eigenvalues → dominant patterns

Small eigenvalues → less important details

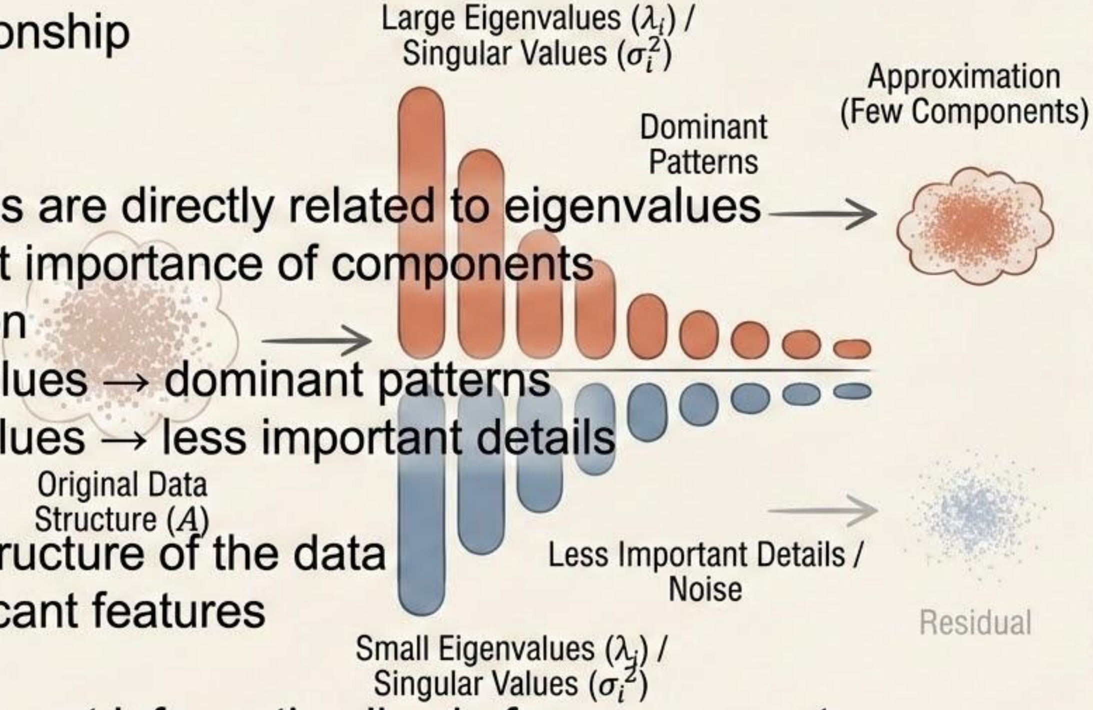
🎯 Purpose

Understand structure of the data

Identify significant features

✅ Outcome

Confirms that most information lies in few components



🔢 Key Relationship:

$\sigma_i^2 = \lambda_i$ of $A^T A$

✅ Outcome:

Confirms that most information lies in few components

Diagonalization & Simplification

👤 **Person 4 — Step 10**

🔢 SVD Representation

$A=U\Sigma V^T$

- Σ is a diagonal matrix
- Contains independent components

💡 Key Idea

System is transformed into independent directions

Each component contributes separately

🎯 Purpose

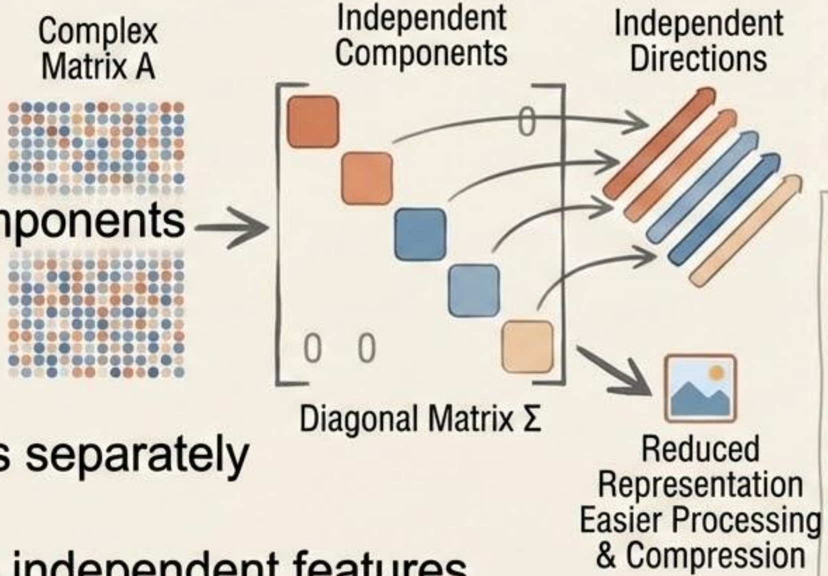
Simplify complex matrix into independent features

Enable efficient storage and computation

✅ Outcome

Reduced representation of image

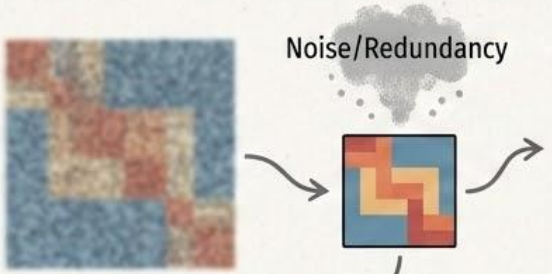
Easier processing and compression



🔢 Key Idea: $A=U\Sigma V^T$ turns complex system into independent components.
Outcome: Enables efficient storage & computation.

From Compression to Insight

💡 Beyond Compression



- Low-rank representation captures essential features
- Removes noise and redundancy

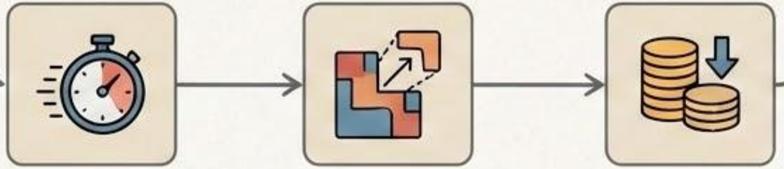
🚀 Future Possibilities

Image classification

Anomaly detection (medical diagnosis)

Pattern recognition in scans

🔍 What This Enables



Faster image processing

Feature extraction for analysis

Reduced computational cost

🎯 Key Idea

Compressed representations can be used as features for downstream predictive tasks

Compressed Representation

Downstream Predictive Tasks

Results Summary

📊 Key Results

- Selected components: $k = 50$ (out of 512)
- Compression ratio: ~10× reduction

🔍 Performance

- Retained >95% of total information (energy)
- Minimal visual loss after compression

💡 Key Observation

Most important information is contained in a small number of components

☑️ Outcome

Efficient compression with high-quality reconstruction

Quality Metrics

Evaluation Metrics

- PSNR (Peak Signal-to-Noise Ratio)
- Measures reconstruction quality
- Higher value → better quality

Compression

- SSIM (Structural Similarity Index)
- Measures structural similarity
- Value closer to 1 → better match

Results

- Average PSNR: ~33 dB
- Average SSIM: ~0.96

Interpretation & ✅ Conclusion

- High PSNR → low reconstruction error
- High SSIM → structure preserved
- Compressed images retain both numerical accuracy and visual structure

Conclusion

✅ What We Achieved

- Implemented a complete linear algebra pipeline
- Successfully compressed medical images using SVD

💡 Key Insights

- Images contain significant redundancy
- SVD identifies and retains important components
- High compression achieved with minimal quality loss

📐 Mathematical Strength

Used concepts:
- Matrix representation
- SVD
- Projection
- Eigenvalues
- Least squares

🎯 Final Takeaway

Linear algebra enables efficient and accurate compression of medical images by removing redundancy while preserving critical information

Applications

🏥 Healthcare Applications

- Telemedicine
  → Faster transmission of medical images
- Cloud Storage
  → Reduced storage cost
- Medical Archives
  → Efficient long-term storage

📱 Technology Applications

- Mobile Diagnostics
  → Efficient processing on low-resource devices
- Image Processing Systems
  → Faster computation and analysis

🔍 Future Scope

- Image classification
- Anomaly detection
- Pattern recognition in medical scans

💡 Key Idea

Efficient compression enables scalable and intelligent medical systems

Thank You

🙏 Thank You & Questions

- ❓ Questions?

💬 Discussion

We're happy to discuss and answer your questions

👥 Team

- Abhinav Agraharam: PES1UG24CS001
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